

Justin Jiang

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EDUCATION

Cornell University

M.Eng. in Computer Science

Ithaca, NY

Expected May 2027

Stony Brook University

B.S. in Computer Science

Stony Brook, NY

May 2026

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL, Bash, HTML/CSS, Go, Rust, C#

Frameworks & Libraries: React, Node.js, Express, Next.js, Django, Tailwind CSS, LangChain, Pandas, NumPy, scikit-learn, Matplotlib, RESTful APIs, .NET

Databases: MongoDB, PostgreSQL

Tools: Docker, Redis, Prometheus, Grafana, Git, AWS

Concepts: Agile, CI/CD, Software Engineering, Design Patterns, Distributed Systems, Microservices

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Web Surfing Studios

September 2026 – Present

Remote

- Contributing to .NET backend services and cloud infrastructure, spanning API development, AWS deployment, and Terraform-based infrastructure-as-code

Research Assistant

Gigacity, Cornell University

September 2026 – Present

Ithaca, NY

- Analyzing SGLang/vLLM scheduling limitations to inform budget-aware scheduling for dynamically expanding agent call trees in real-time transportation-response systems

Research Assistant

COMPAS Lab, Stony Brook University

January 2026 – May 2026

Stony Brook, NY

- Built a Python CLI tool automating local setup of a distributed genomic search system, provisioning isolated coordinator/worker environments and containerized MongoDB instances with a single command in place of a fragile multi-step manual workflow
- Diagnosed and patched a data corruption bug in the system's probabilistic indexing structure caused by integer underflow during high-load merge operations

Machine Learning Fellow

Break Through Tech AI

May 2025 – April 2026

New York, NY

- Built TaskPilot, a Slack-integrated productivity agent using Gemini 2.5 Flash, implementing confidence-based clarification-trigger logic, Pydantic schema validation, and thread-scoped session management as part of an agentic AI specialization
- Built a Streamlit dashboard under Snowflake engineers, training classification models on weather, hospital, and social vulnerability data to flag medical-desert risk across Arizona

Systems Programming Teaching Assistant

Stony Brook University

August 2024 – December 2024

Stony Brook, NY

- Led weekly recitations and office hours on C programming and systems debugging for 100+ students; collaborated with course staff to design problem sets and exam questions

PROJECTS

Distributed Key-Value Store | *Java, RPC, Fault Tolerance, Concurrency*

September 2026 – Present

- Building a fault-tolerant, sharded, transactional key-value store modeled on production systems like Google Spanner
- Implemented exactly-once RPC delivery over an unreliable network
- Currently extending to Paxos-based replication across multiple nodes

Discord Clone | *TypeScript, PostgreSQL, WebSockets, Docker, Redis* | [GitHub](#)

March 2026 – May 2026

- Deployed a real-time messaging platform on an 8-node Docker Swarm cluster, supporting 4K peak concurrent connections at 240+ req/s
- Identified and resolved lock contention between GIN index updates for full-text search and message insert paths by decoupling search indexing onto async background workers, dropping insert latency from 200ms to <1ms
- Built a distributed Redis caching layer for high-frequency read-state updates, preventing PostgreSQL connection saturation under load